

BARRETT KODANDARAM

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EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Master of Science in Computer Science - 4.0/4.0

Jan 2025 - Dec 2026

Relevant Coursework: Algorithms, Operating Systems, Deep Learning, Modern Parallel and Distributed Computing

EXPERIENCE

Graduate Research Assistant

Sep 2025 - Present

APECS Lab, Stevens Institute of Technology

Hoboken, NJ

- Engineered a fused CUDA/C++ kernel for distributed long-context LLM inference, reducing per-token latency by 9% on dense models and 2% on Mixture-of-Experts (MoE) models.
- Profiled decode latency in context-parallel workloads deployed across 32 A100 GPUs on Argonne National Lab's Polaris supercomputer (TensorRT-LLM, Nsight Systems), motivating the fused kernel design.
- Built and maintained the lab's Apptainer/Docker images through 3 breaking Polaris system updates by remapping host libraries, averting 2 months of downtime.
- Benchmarked a lab-developed GPU collectives library against NVIDIA NCCL, measuring a 14% reduction in communication time during pre-training.

PROJECTS

Lean-Infer | Python, PyTorch, Prometheus, Grafana, Docker Compose, GCP

Jul 2026

- Built an LLM inference engine in PyTorch that fits 20% more concurrent requests on a single cloud GPU (L4 on GCP), using a vLLM-style paged KV cache and a continuous-batching scheduler.
- Achieved 8x decode throughput (1400 tokens/sec with Qwen3-0.6B) at 85% memory-bandwidth utilization via FlashAttention and CUDA Graph capture.
- Instrumented the engine with Prometheus metrics and Grafana dashboards for observability, localizing the bottlenecks each optimization targeted.

Resilient Distributed Rideshare | Go, gRPC, PostgreSQL, Kafka, Redis, Docker Compose

Mar 2026

- Built a fault-tolerant rideshare backend in Go that preserves data consistency when a microservice crashes mid-request, using idempotent consumers and a transactional outbox (PostgreSQL, Kafka).
- Verified automatic recovery of 100% of orphaned rides across 5 crash scenarios spanning every ride-lifecycle stage, using a fault-injection script that kills services mid-transaction.

Before You Sign | TypeScript, Node.js, Express.js, MongoDB, Zod

May 2026

- Implemented an ETL pipeline with TypeScript, MongoDB and Zod that ingests up to 50K NYC housing violation records per run and alerts subscribers within 5 minutes of publication.
- Owned the type-safe data layer for a 5-person team, designing 7 Zod-validated Mongoose models and documenting the schema conventions the team built against.

TECHNICAL SKILLS

Core Languages: Python, Go, C, C++, SQL, JavaScript, TypeScript, Bash/Shell

AI/ML Systems: PyTorch, CUDA, Triton, vLLM, TensorRT-LLM, Nsight, HPC

Backend & Data: PostgreSQL, MongoDB, Redis, Kafka, gRPC, REST APIs

Cloud, Infra & Observability: AWS (ECS, EC2, S3), GCP (GCE), Docker, Linux, Git, Prometheus, Grafana